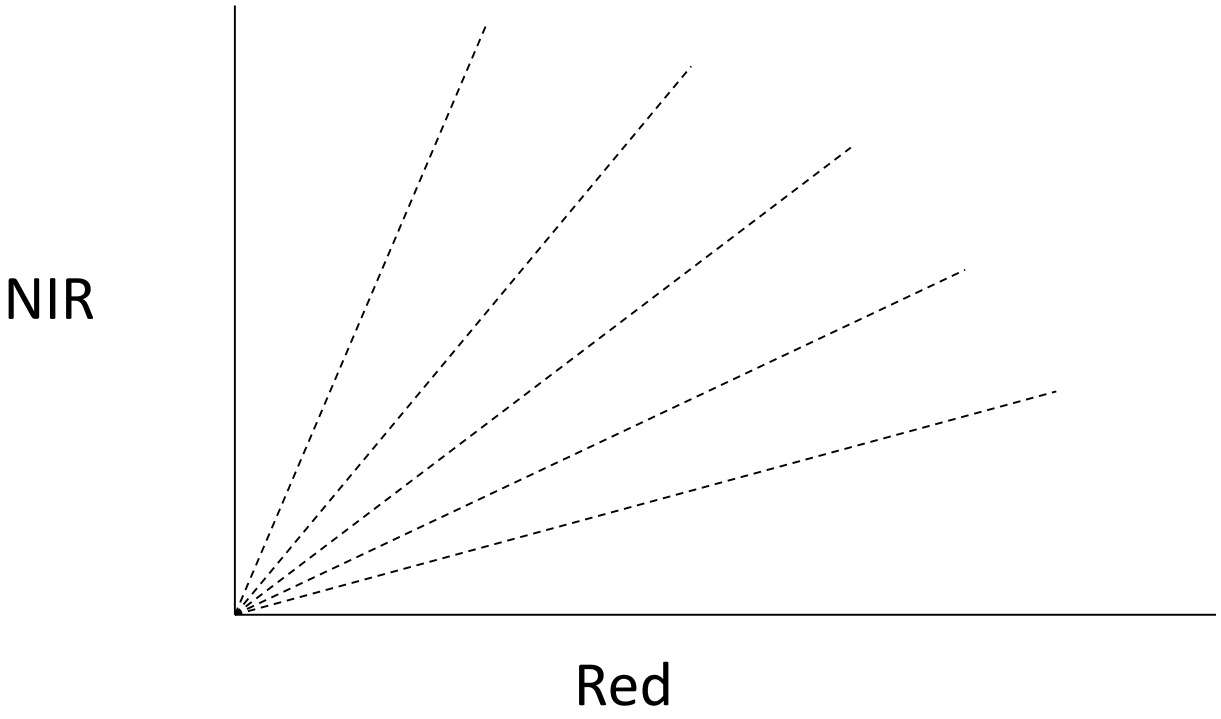
Vegetation Index (VI)

Vegetation Indices

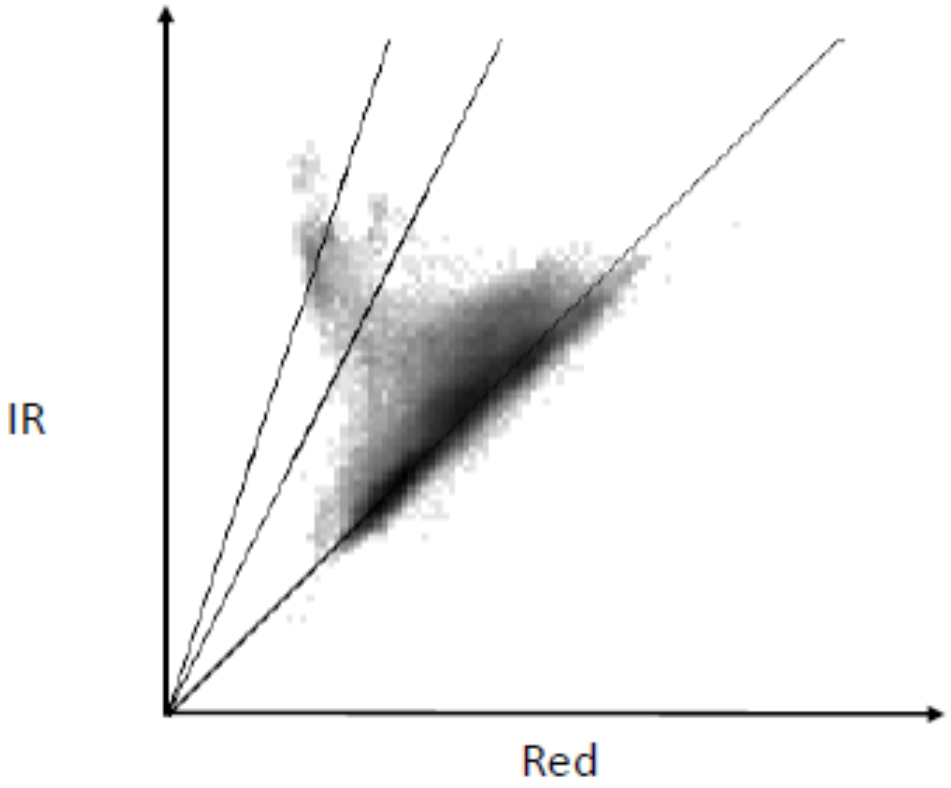
- Main categories:
  - Ratio indices
  - Perpendicular indices
- Ratio indices
  - Ratio Vegetation Index (RVI)
  - Normalized Difference Vegetation Index (NDVI)
- Perpendicular indices
  - Perpendicular Vegetation Index (PVI)
  - Soil Adjusted Vegetation Index (SAVI)

Ratio indices

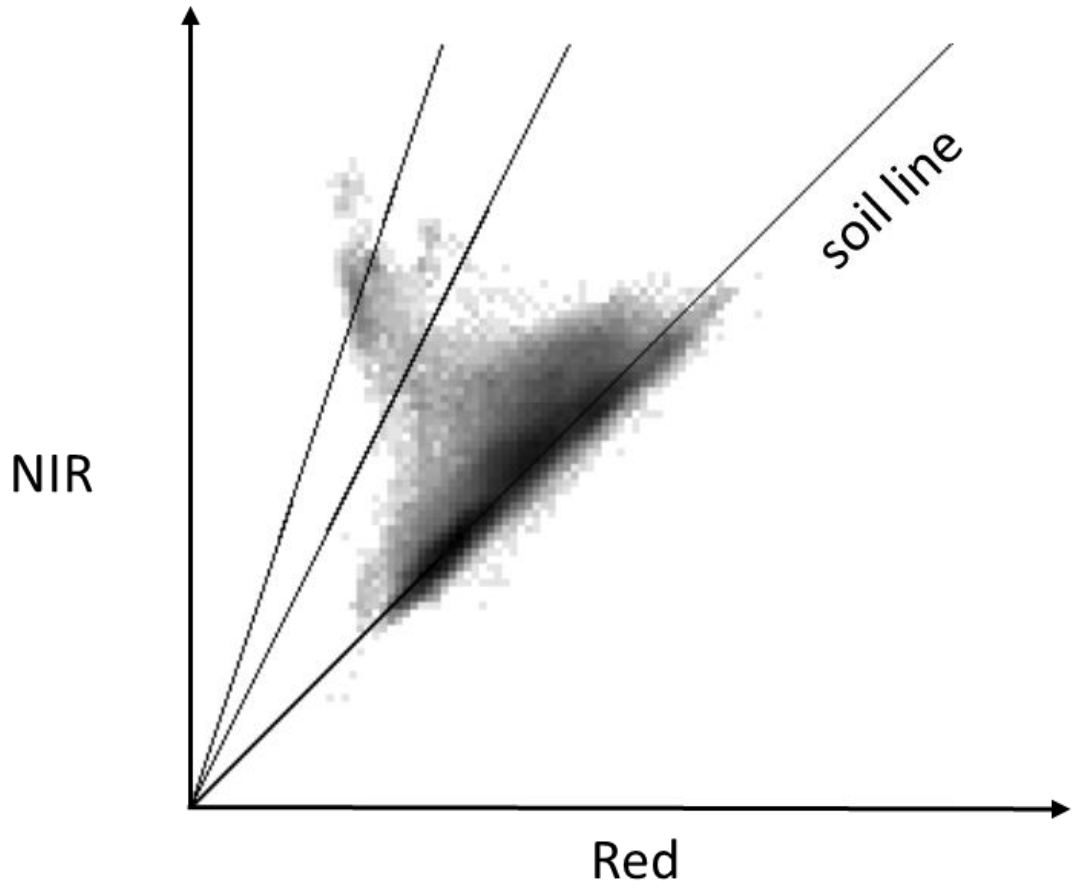
- Both RVI and NDVI basically measure the slope of the line between the origin of red-NIR space and the red-NIR value of the image pixel



DN Scattergram for Red versus IR Bands



Soil Line



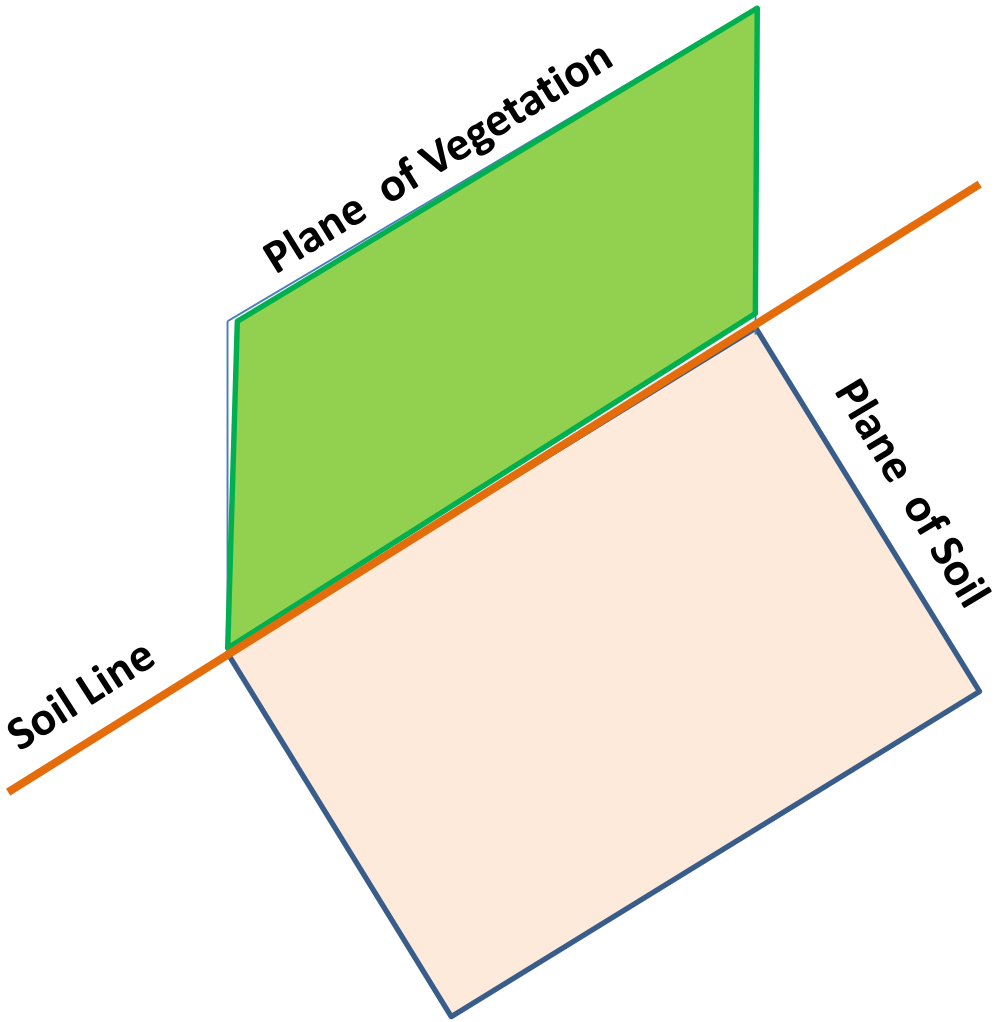
- In RED-NIR plot, lower right boundary is taken to be formed by pixels containing only bare soil, and is referred to as the Soil Line
- It is used as a reference point in most vegetation related studies

Soil Line

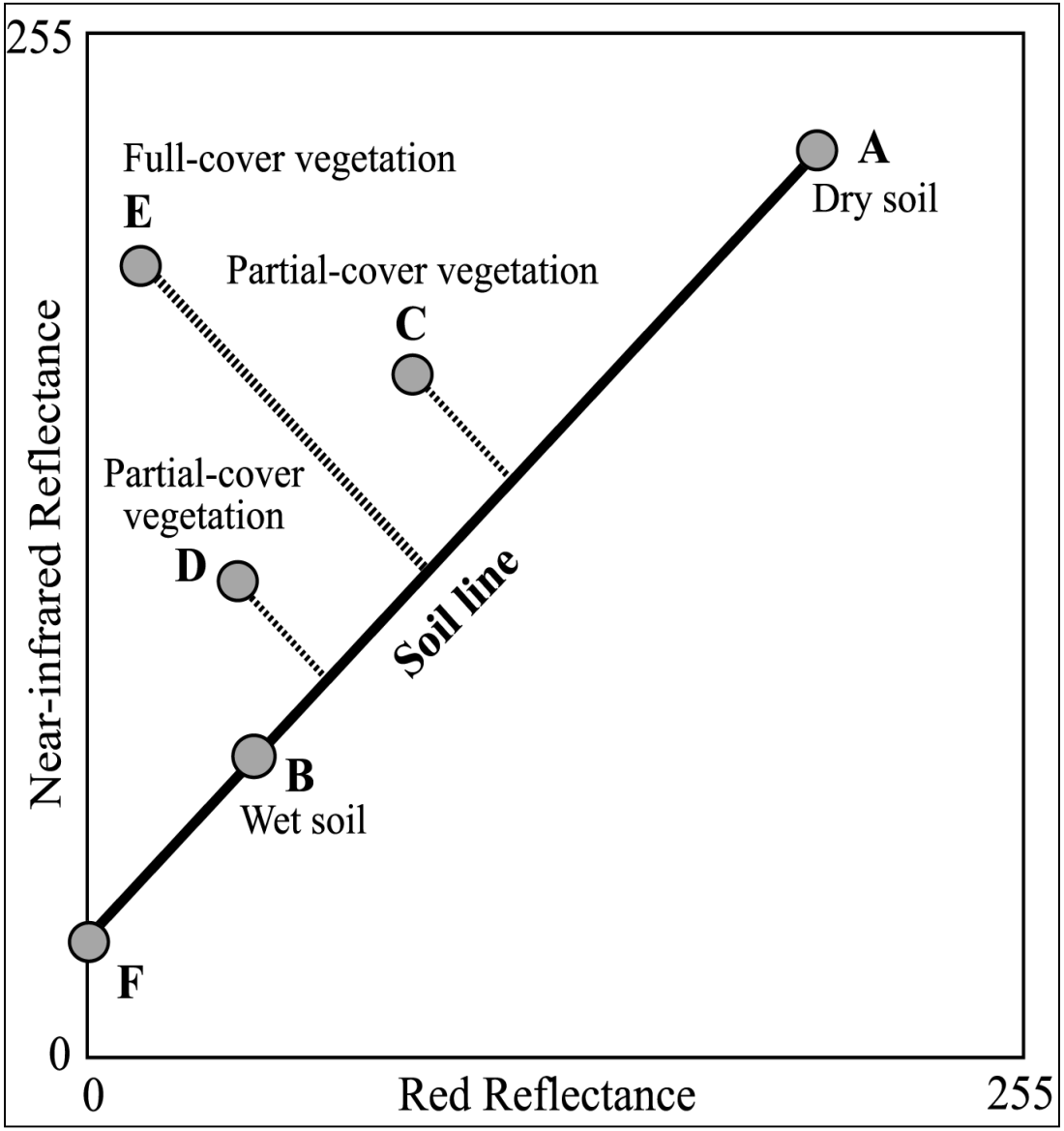


- Soil Line is recognized that the spectral response of living vegetation will always have a consistent relationship to the line

Soil Line



Soil Line



- Most vegetation indices rely in the existence of a "Soil Line" in red and NIR wavelength space
- There is a principal axis of soil spectral variation extending outward from the origin (red and NIR spectral space) with increasing brightness
- Most of the soil spectra fall on or close to the soil line

Soil Line

- Soil line will be different for different areas (soil types)
- Soil line will vary for different NIR and red band passes
- Truly valid way of making use of a vegetation index which uses a soil line is to compute the soil line for each image
- Assumption that all of the bare soil spectra in a single image form a line may also be inaccurate

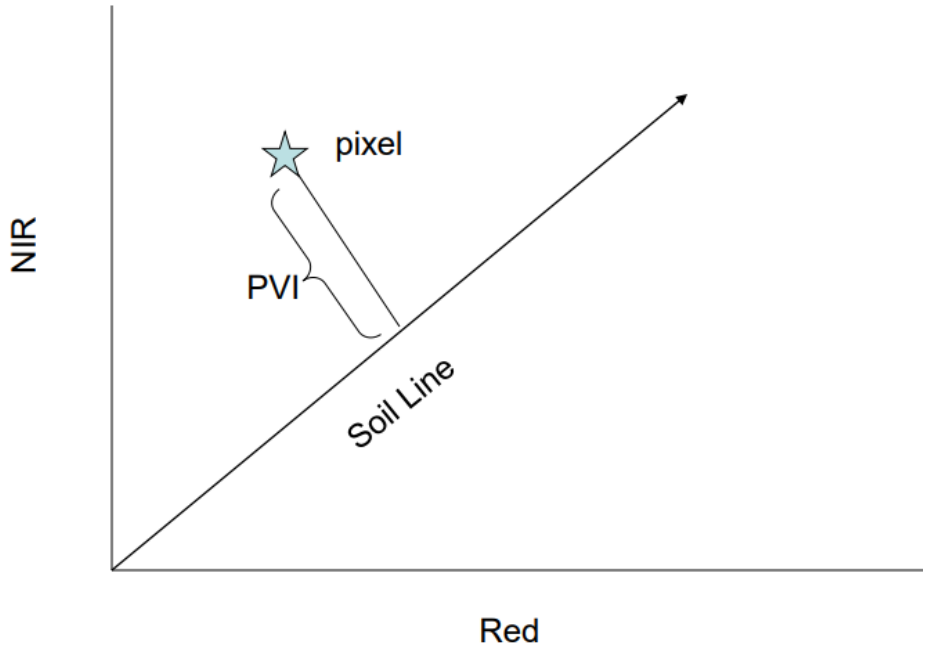
Soil Line

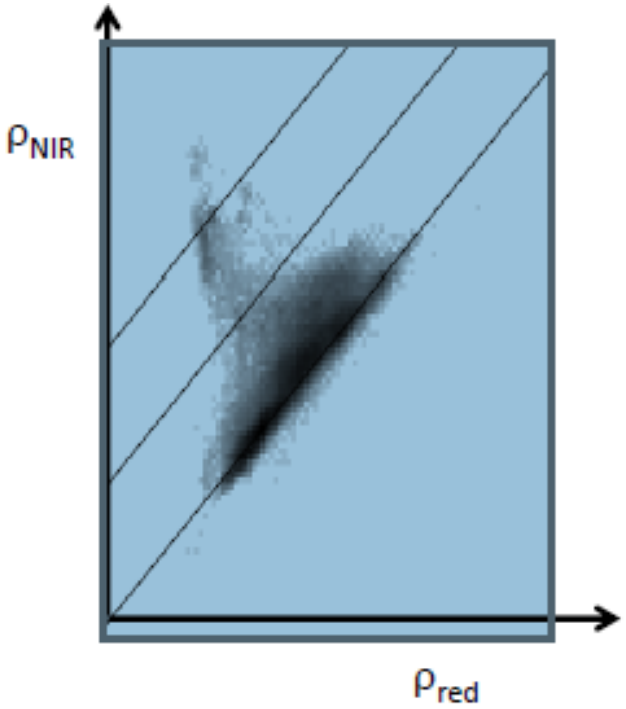
- The experimental data revealed that it is not feasible to define a single, universally applicable soil brightness line, as local variations in soil types lead to spectral variations
- However, these variations were found to be minor in the red and
- infrared regions

Perpendicular Vegetation Index (PVI)

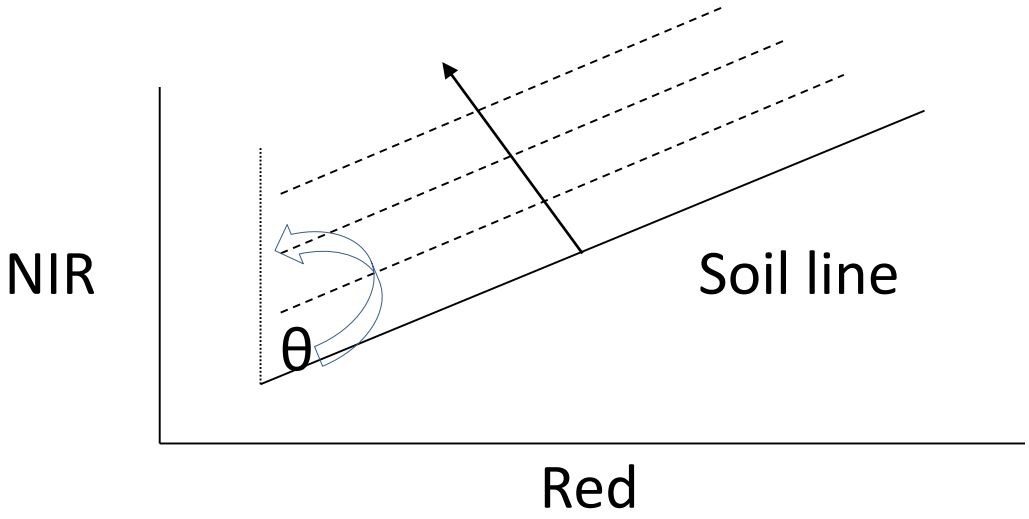
PERPENDICULAR VEGETATION INDEX (PVI)

- PVI assumes that the perpendicular distance of the pixel from the soil line is linearly related to the vegetation cover
- The iso-vegetation lines (lines of equal vegetation) would all be parallel to the soil line





Indices Using the Soil Line -PVI



- This index is calculated as follows:

$$PVI_{(NIR\ red)} = - \text{Sin } \theta \text{ (NIR) x Cos } \theta \text{ (red)}$$

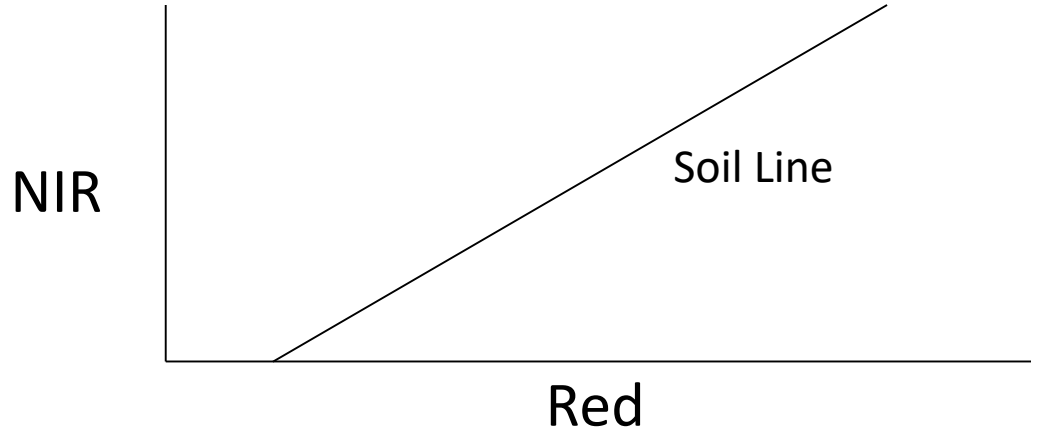
- where (NIR) is the near-infrared reflectance, (red) is the red reflectance and (θ) is the angle between the soil line and the near-infrared axis

Richardson and Wiegand (1977)

G uyon et al. (2011) used a more generic PVI to monitor leaf phenology in deciduous broadleaf forests

$$PVI = \frac{(NIR - aRed - b)}{\sqrt{1 + a^2}}$$

Indices Using the Soil Line -PVI

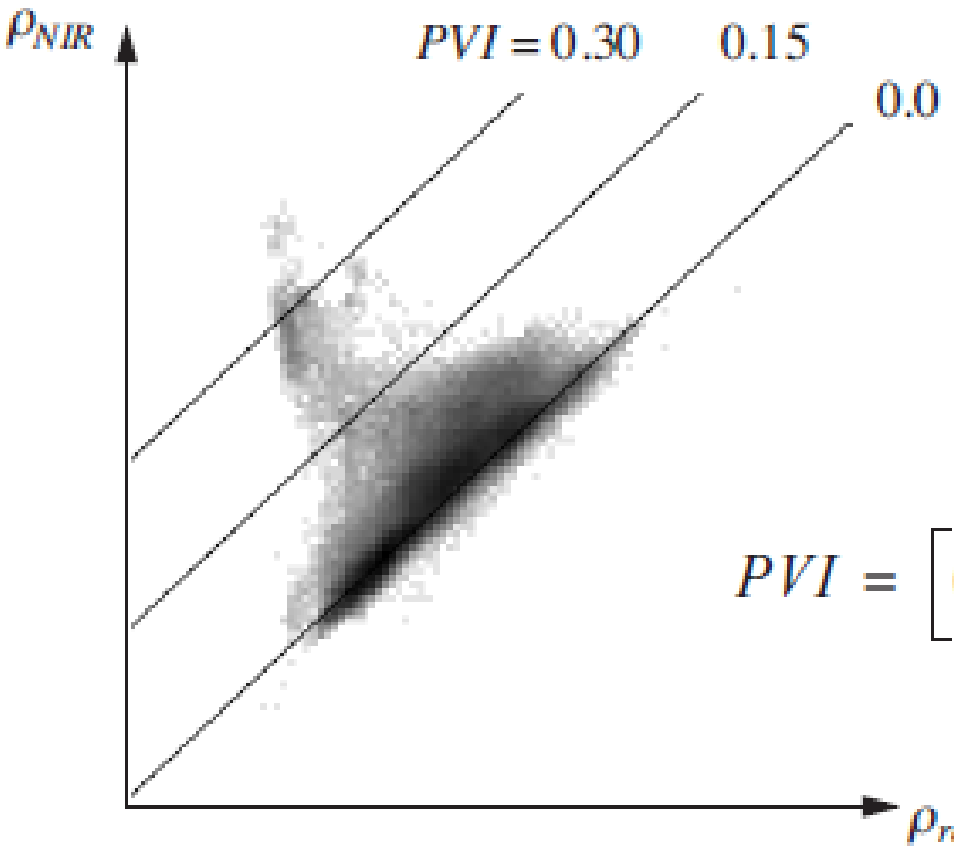


Guyon et al. (2011) used a more generic PVI to monitor leaf phenology in deciduous broadleaf forests

$$\mathrm{PVI} = \frac{(NIR - aRed - b)}{\sqrt{1 + a^2}}$$

Where: a and *b* are the slope and the intercept of the soil line, respectively

Perpendicular Vegetation Index

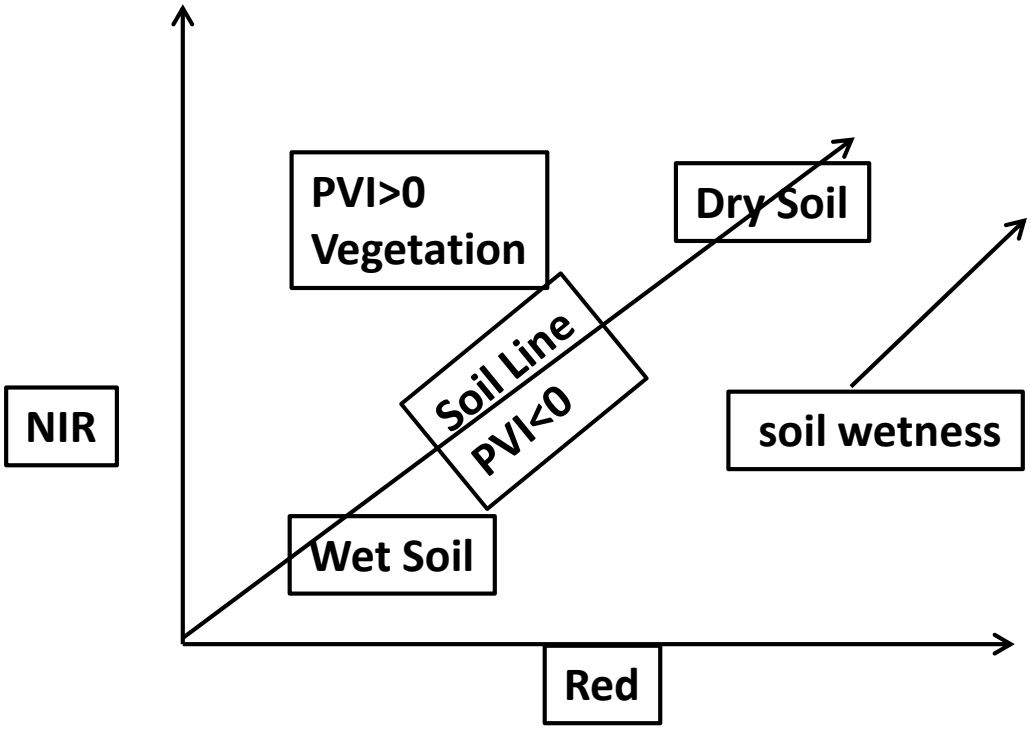


$$PVI = \left[\left(\rho_{red}^{soil} - \rho_{red}^{veg} \right)^2 + \left(\rho_{NIR}^{soil} - \rho_{NIR}^{veg} \right)^2 \right]^{1/2}$$

PVI as measure of the perpendicular distance of a pixel (in spectral space) from the soil brightness line

Perpendicular Vegetation Index

- A plot of radiance measured in the red band against in the near-infrared for a partly-vegetated area



Soil-Adjusted Vegetation Index (SAVI)

Soil-Adjusted Vegetation Index (SAVI)

- Vegetation index designed to minimize the effect of soil background
- A transformation technique to minimize soil brightness from spectral vegetation indices involving red and NIR wave lengths

Soil background

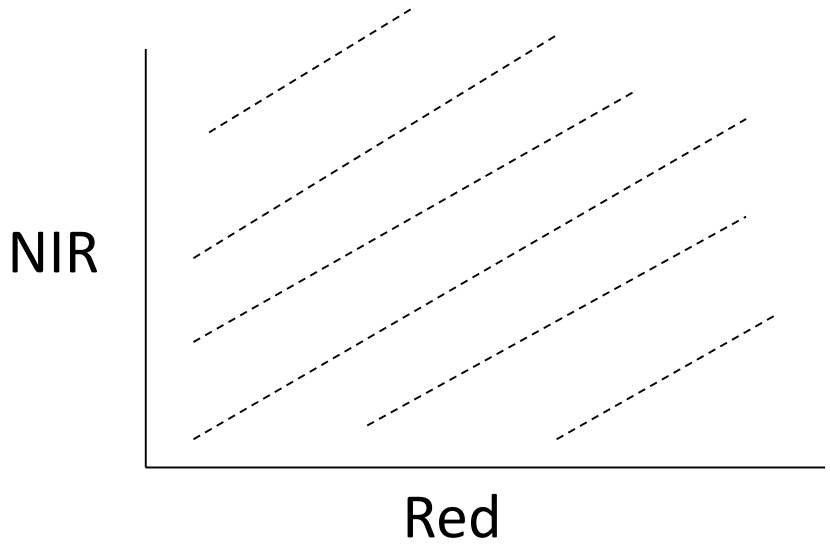


SAVI Images

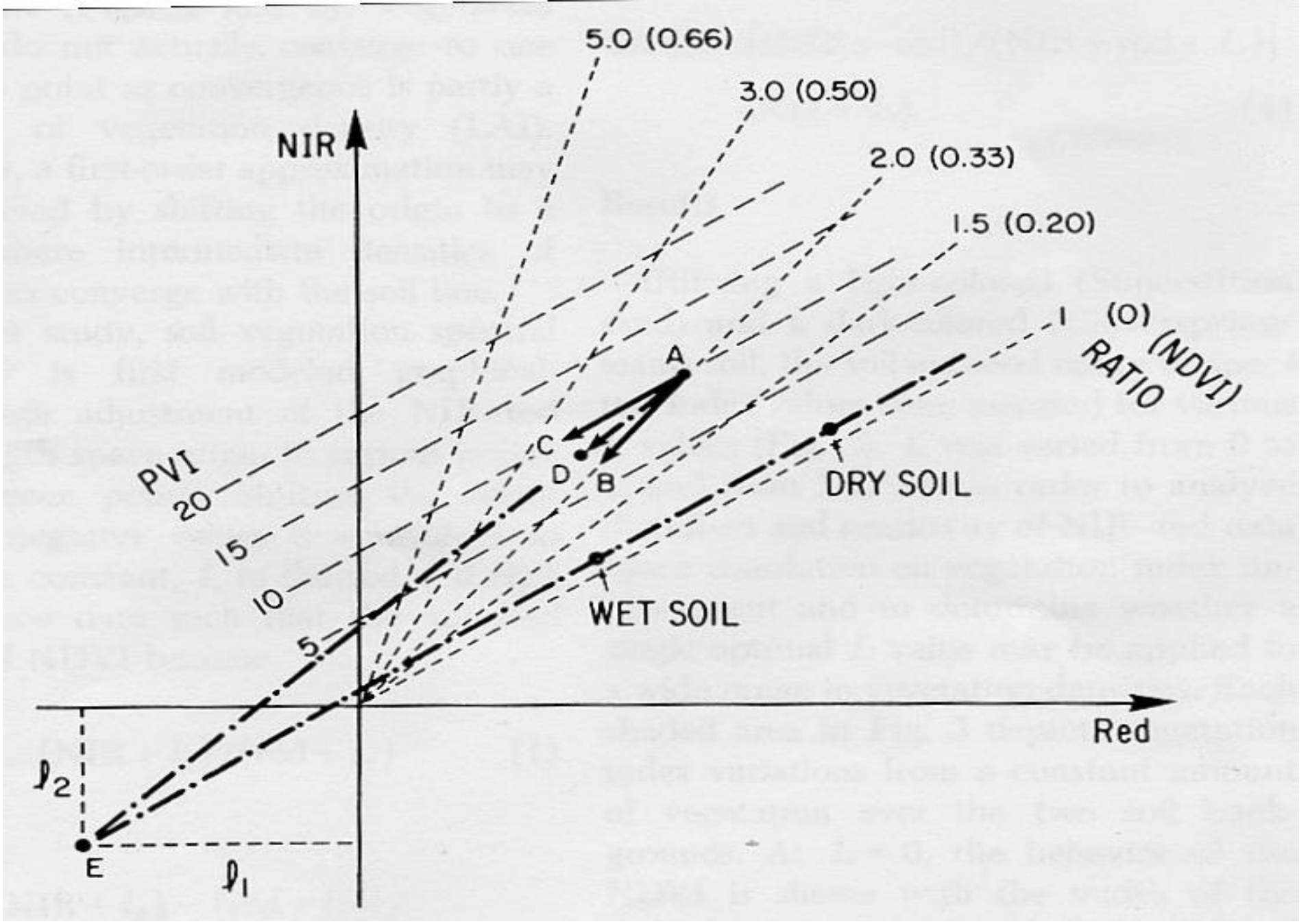
- Vegetation information is difficult to extract when vegetation cover is less than 30%
- In arid and semiarid regions, vegetative cover may always be this low, but even in agricultural settings plant cover is low when crops are emerging early in the growing season
- The Soil-Adjusted Vegetation Index (SAVI) is a superior vegetation index for low cover environments

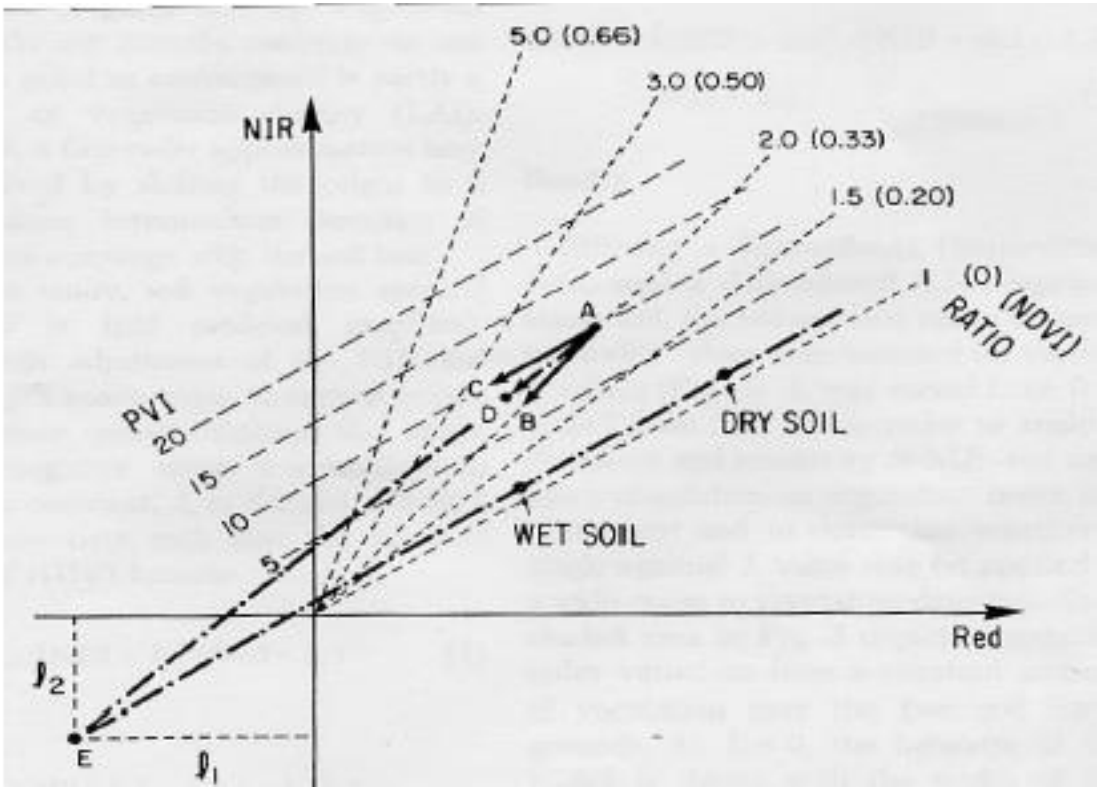
Soil-Adjusted Vegetation Index (SAVI)

- Huete showed evidence that the isovegetation lines (lines of equal vegetation) do not converge at a single point
- Selected the L-factor in SAVI, where lines of a specified vegetation density intersect the soil line in 3rd Quadrant-Negative NIR and Red



Huete (1988)


5.0 (0.66)
3.0 (0.50)
2.0 (0.33)
1.5 (0.20)
NIR
A
C
D
B
PVI
20
15
10
5
WET SOIL
DRY SOIL
1
(0)
(NDVI)
RATIO
Red
l_2
E
l_1



- A partial canopy cover a dry soil background **(A)**
- If the soil background were to become wet,
  - RVI and NDVI to effectively normalized such a background change, the vegetated pixel
  - would have to shift directly towards the origin **(B),** following the isoline of constant RVI and NDVI values
  - PVI would however require the pixel to shift along isoline parallel to the soil line **(C),** so that both the wet and dry soil vegetated pixels maintain a constant PVI value
- They found that vegetation isolines to possess NIR-Red wavelength slopes in between those of ratio and perpendicular based Isolines **(D)**
- The isoline neither converged at the origin (as required by the RVI and NDVI),nor maintained
- Slopes identical with straight parallel line (A PVI requirement) Rather ,they converged in 3rd quadrant **(E)**
-

Soil-Adjusted Vegetation Index (SAVI)

To minimize the effects of soil, adjustment factors (l1,l2) are introduced in SAVI for VI and NDVI

For VI $$\frac{NIR + l_2}{red + l_1}$$

For NDVI $$\frac{(NIR + l_2) - (red + l_1)}{(NIR + l_2) + (red + l_1)}$$

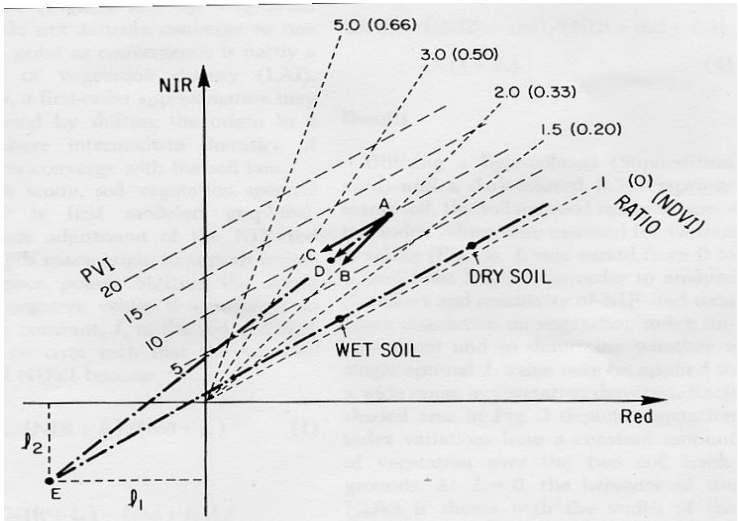


FIGURE 1. Vegetation spectra isolines in NIR–red wavelength space as predicted by the ratio-, normalized difference-, and perpendicular-vegetation indices.

Soil-Adjusted Vegetation Index (SAVI)

- Soil line has slope close to 1, the adjustment factors, l_1 and l_2, would be nearly equivalent
- Shifting the red and NIR data equally (l_1=l_2) and utilizing the NDVI format

$$\frac{NIR - red}{NIR + red + L} \qquad \frac{(NIR + l_2) - (red + l_1)}{(NIR + l_2) + (red + l_1)}$$

Where $L=l_1+l_2$

Thus, SAVI would only involve an addition of a constant L, to the denominator of the NDVI equation

L is a constant that is empirically determined to minimize the vegetation index sensitivity to soil background reflectance variation

Soil-Adjusted Vegetation Index (SAVI)

However, in order to maintain the bounded conditions of the NDVI equation (NDVI can vary from -1 to +1), a multiplication factor (1+L) is needed

$$SAVI = \frac{NIR - red}{NIR + red + L} \times (1 + L)$$

L = Function of the shifted origin correction factor

$$SAVI = (1 + L)\frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red} + L}$$

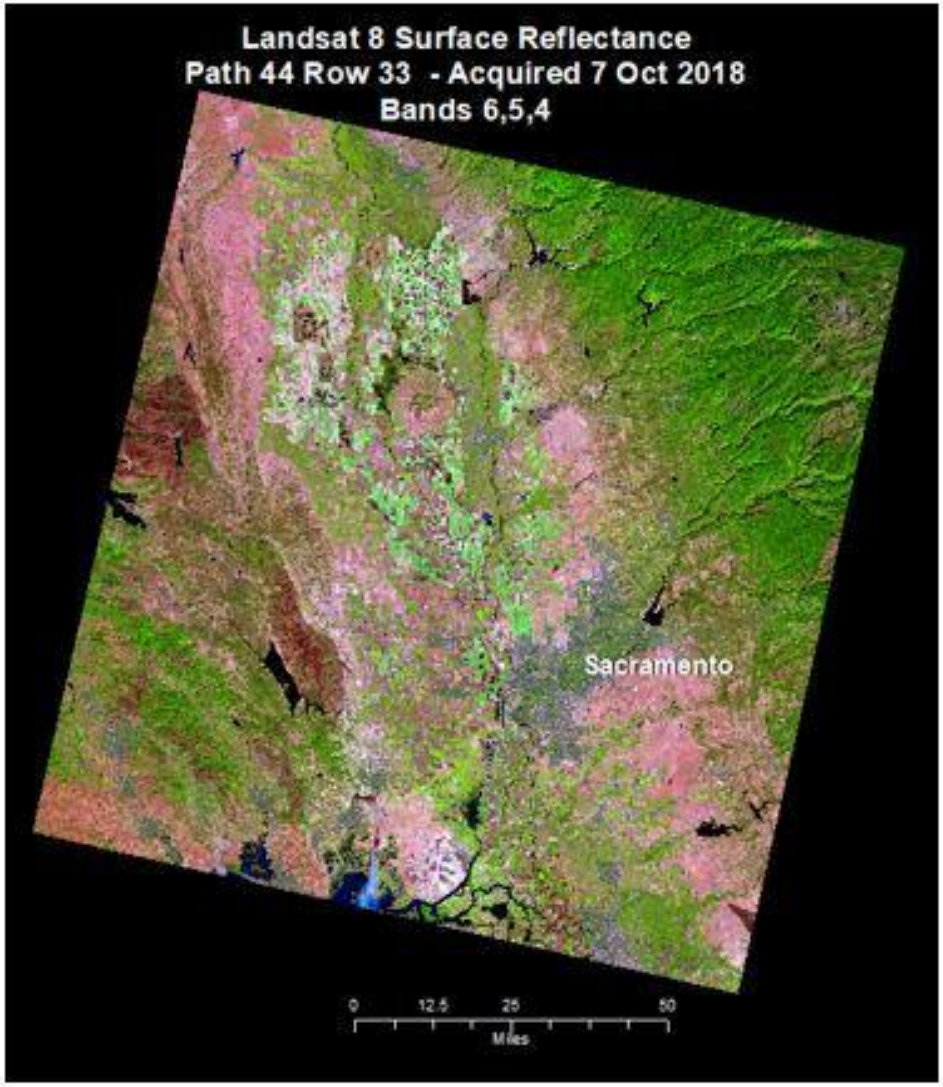
ρ_{NIR} = reflectance in near-infrared band
ρ_{Red} = reflectance in red band
L = 0.5

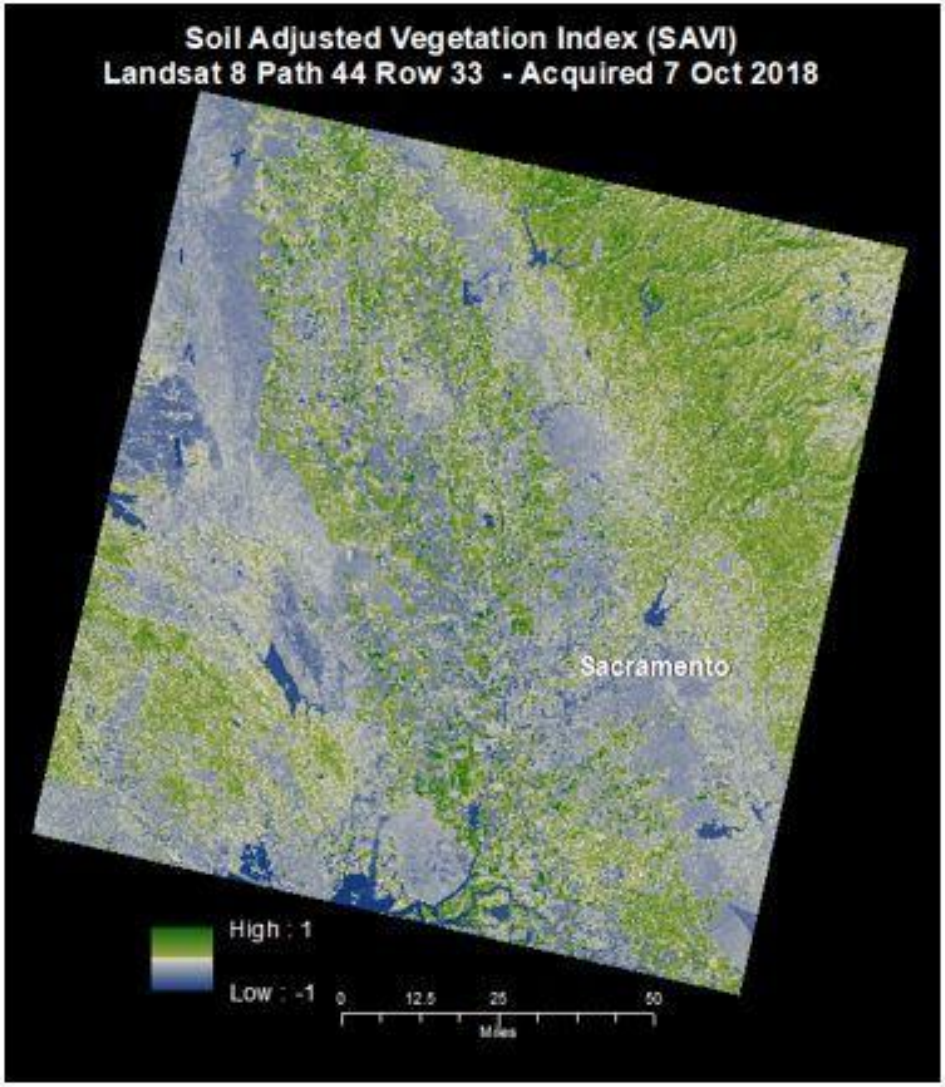
SAVI

- For high vegetation cover, the value of L is 0.0, and L is 1.0 for low vegetation cover
- For intermediate vegetation cover L=0.5, and that is the values which is most widely used
- The SAVI has a range identical to the of NDVI (-1 to +1)
- Huete (1988) suggested that SAVI takes on both the aspects of NDVI and PVI

SAVI

- Elvidge and Chen (1995) found that SAVI and PVI consistently provided better estimates of percent green cover than did NDVI or RVI
- They also found that there was a steady improvement in all of these vegetation indices as narrower and narrower bands were used for the near-infrared and red reflectances, **with SAVI being the best index at the very narrowest bandwidth**


Landsat 8 Surface Reflectance
Path 44 Row 33 - Acquired 7 Oct 2018
Bands 6,5,4
Sacramento
0 12.5 25 50
Miles


Soil Adjusted Vegetation Index (SAVI)
Landsat 8 Path 44 Row 33 - Acquired 7 Oct 2018
Sacramento
High : 1
Low : -1
0 12.5 25 50
Miles

L=0.5

TSAVI

- A further development of this concept is the Transformed SAVI (TSAVI), Baret and Guyot, 1991), defined as

$$TSAVI = a(NIR - aR - b)/[a(NIR - b) + 0.08(1 + a^2)]$$

- Where a and b are, respectively, the slope and intercept of the soil line
- and the coefficient value 0.08 (adjustment factor) has been adjusted to minimise soil effects,
- NIR = pixel values from the near infrared band,
- R = pixel values from the red band

MSAVI

- Qi et al. (1994a) further developed a vegetation index which is basically a version of SAVI
- They referred to this index as the Modified Soil Adjusted Vegetation Index or MSAVI
- The factor L is given by the following expression:

$$MSAVI = \frac{\left\{ 2NIR + 1 - \left[(2NIR + 1)^2 - 8(NIR - red) \right]^{0.5} \right\}}{2}$$

Enhanced Vegetation Index (EVI)

- EVI is similar to Normalized Difference Vegetation Index (NDVI) and can be used to quantify vegetation greenness.
- However, EVI corrects for some atmospheric conditions and canopy background noise
- It incorporates an "L" value to adjust for canopy background, "C" values as coefficients for atmospheric resistance, and values from the blue band (B).

$$EVI = G \times \frac{(NIR - Red)}{(NIR + C_1 \times Red - C_2 \times Blue + L)}$$

- In Landsat 8-9,

EVI = 2.5 * ((Band 5 – Band 4) / (Band 5 + 6 * Band 4 – 7.5 * Band 2 + 1))

C1,C2 and L are empirically derived values and for most satellite sensor, 6, 7.5 and are used as standard values

Enhanced Vegetation Index (EVI)

$$EVI = \frac{p * nir - p * red}{p * nir + C_1 p * red - C_2 p * blue + L}$$

?